SE3115: Software Testing and Quality Assurance

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Course Teacher:

M Saeed Siddik,

Assistant Professor, Institute of IT, University of Dhaka

URL: https://sites.google.com/view/saeedsiddik

Learning outcomes

By the end of the course, students will be able to:

1. Explain core testing concepts (fault, failure, oracle) and modern quality strategy: test pyramid vs testing trophy, shift-left and shift-right, testing metrics.
2. Derive test cases by hand using equivalence partitioning, boundary values, decision tables, state models and pairwise testing.
3. State properties and invariants for property-based testing, and explain how fuzzing finds faults.
4. Draw control-flow graphs and compute statement, branch, condition and MC/DC coverage; explain why mutation score is a stronger measure than coverage.
5. Explain unit, integration, contract and end-to-end testing, and choose the right level for a given risk in a microservice system.
6. Describe how a CI/CD pipeline runs tests and gates releases, and analyses causes of flaky tests.
7. Explain security testing (SAST, DAST, dependency scanning, fuzzing) and map threats to the OWASP Top 10.
8. Explain load testing, SLOs, observability, canary releases, feature flags and chaos engineering.
9. Explain why machine-learning systems lack exact oracles, and apply metamorphic, robustness and fairness testing to them.
10. Design evaluations for LLM, RAG and agent-based systems, and plan red-team tests against prompt injection and jailbreaks; relate them to the EU AI Act and NIST AI RMF; critically review AI-generated tests.

Weekly schedule

Weeks 1 to 7 build test-design skills students can practice on paper; weeks 9 and 10 survey modern delivery practice; weeks 11 to 13 are a dedicated block on testing AI-based intelligent systems. The midterm falls in week 8.

Week	Topic	Key content	Reference Books
1	Foundations of testing	Fault, error, failure; the oracle problem; limits of testing; verification vs validation; testing vs debugging	Textbook-1: Chapter 1, 2; Textbook-2: Chapter 1, 2; Textbook-3: Chapter 1;

Week	Topic	Key content	Reference Books
2	Testing in modern development	Test levels, V-model vs agile and DevOps, test pyramid vs testing trophy, shift-left and shift-right, DORA metrics	Textbook-2: Chapter 1, 2; Textbook-3: Chapter 3;
3	Black-box design I	Equivalence partitioning, boundary value analysis	Textbook-2: Chapter 5, 6; Textbook-3: 4
4	Black-box design II	Decision tables, cause-effect graphs, state transition testing, pairwise and combinatorial testing	Textbook-2: Chapter 7; Textbook-3: 4
5	White-box testing I	Coverage Criteria, Control-flow graphs, statement and branch coverage, cyclomatic complexity, basis paths	Textbook-2: Chapter 8; Textbook-3: 5
6	White-box testing II	Graph Metrics, Data Flow testing, Mutation Testing.	Textbook-2: Chapter 8; Textbook-3: 5
7	Unit testing principles and TDD	Test structure (arrange-act-assert), test doubles, TDD cycle, test smells, property-based testing and fuzzing; Software Quality Management	Textbook-3: 13 Papers on test structures, test smells.
8	Midterm exam	Weeks 1 to 7	—
9	Test Case Prioritization	Regression Testing, Test suite minimization, Test case prioritization, Fault detection metrics	Slide will be provided
10	Integration, CI/CD and end-to-end testing	Integration strategies, API and consumer-driven contract testing for microservices, pipeline quality gates, flaky tests, E2E, visual and accessibility (WCAG 2.2) testing	Choose test levels and pipeline gates for a microservice architecture diagram
11	Security, performance and reliability	OWASP Top 10, SAST, DAST, dependency scanning; load testing and SLOs; observability, canary releases, feature flags, chaos engineering	Case study: Log4Shell and the CrowdStrike outage (July 2024)
12	Testing machine-learning systems	Why ML breaks classic testing (no exact oracle, data-driven behaviour, non-determinism); data validation; evaluation metrics and test sets; metamorphic testing; robustness and adversarial examples; fairness and bias testing; drift monitoring; ML Test Score	Design metamorphic relations for an image classifier; case study: COMPAS bias analysis

Week	Topic	Key content	Reference Books
13	Testing LLM and generative AI systems	Evaluation datasets and golden answers, LLM-as-judge and its limits, hallucination and factuality checks, RAG evaluation (retrieval quality, faithfulness); Prompt regression testing, testing AI agents (tool use, multi-step tasks, sandboxes)	Build an evaluation rubric for a customer-support chatbot; case study: Air Canada chatbot ruling (2024)
14	Safety, security and governance of AI systems; AI as a testing tool	Prompt injection, jailbreaks and red-teaming (OWASP Top 10 for LLM Applications); scenario- and simulation-based testing of autonomous systems; EU AI Act, NIST AI RMF and ISO/IEC 42001 testing duties; reviewing AI-generated tests	Red-team a described LLM app on paper; critique a set of AI-generated unit tests; case study: Uber self-driving crash (2018)

Grading:

Component	Weight
Final exam	50%
Midterm exam	25%
Assignments and Presentation	15%
Attendance and participation	10%

Textbooks, tools and resources

Main textbook

- Text Book 1: Myers, Glenford J., Corey Sandler, and Tom Badgett. The art of software testing. John Wiley & Sons, 2011.
- Text Book 2: Jorgensen, Paul C. Software testing: a craftsman's approach. Auerbach Publications, 2013.
- TextBook 3: Chauhan N., Software Testing: Principles and Practices, 2nd Edition, Oxford University Press, 2010

Reference books

- Maurício Aniche, *Effective Software Testing: A Developer's Guide*, Manning, 2022.
- Titus Winters, Tom Manshreck and Hyrum Wright (eds.), *Software Engineering at Google*, O'Reilly, 2020 -chapters 11 to 14 on testing at scale (free online).
- Nicole Forsgren, Jez Humble and Gene Kim, *Accelerate*, IT Revolution, 2018 , DORA metrics and delivery performance.
- Casey Rosenthal and Nora Jones, *Chaos Engineering*, O'Reilly, 2020.
- Chip Huyen, *Designing Machine Learning Systems*, O'Reilly, 2022 -data validation, evaluation and monitoring of ML (week 11).
- Chip Huyen, *AI Engineering*, O'Reilly, 2025 - evaluating LLM, RAG and agent applications (week 12).
- J. M. Zhang, M. Harman, L. Ma and Y. Liu, "Machine Learning Testing: Survey, Landscapes and Horizons," *IEEE Transactions on Software Engineering*, 2020 - the standard survey for week 11.
- E. Breck et al., "The ML Test Score: A Rubric for ML Production Readiness," IEEE Big Data, 2017.
- OWASP Web Security Testing Guide and OWASP Top 10 for LLM Applications (online).
- NIST AI Risk Management Framework 1.0 (2023) and the EU AI Act (2024), for week 13.